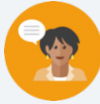


3.H1 Plotting and Analyzing Residuals

Lesson Introduction

 Benchmark	MA.912.DP.2.5 Given a scatter plot that represents bivariate numerical data, assess the fit of a given linear function by plotting and analyzing residuals.		 Learning Targets	- I can create a residual plot to assess the fit of a given linear function for bivariate data. - I can determine whether a function is an appropriate fit for the data by analyzing a residual plot.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- How do you create a residual plot to assess the fit of a given linear function for bivariate data? - How do you determine whether a function is an appropriate fit for the data by analyzing a residual plot?			
 Lesson Narrative	In this lesson, students learn how to create a residual plot to assess the fit of a given linear function for bivariate data, as well as determine whether a function is an appropriate fit for the data by analyzing the residual plot. Students begin by working together to create a residual plot given a data set representing a real-world situation. They will then explore how to determine the fit of a function by analyzing residuals.			
 Component Summary	3.H1.1 Warm-Up (~5 min.)	Which One Doesn't Belong (<i>SE p. 156</i>)	3.H1.3 Exploration (15 min.)	Explore how to determine the fit of a function by analyzing residuals (<i>SE p. 158</i>)
	3.H1.2 Collaboration (15 min.)	Create a residual plot for a given set of bivariate data (<i>SE p. 157</i>)	3.H1.4 Bring It Together (10 min.)	Formalize how to determine the fit of a function by analyzing residuals (<i>SE p. 159</i>)
	3.H1.5 Wrap-Up (~5 min.)	Ticket Out the Door on analyzing residuals (<i>SE p. 160</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Residual - Outlier <u>Additional Glossary:</u> N/A		N/A	
	Additional Preparation (Optional) Take the data sets in 3.H1.3 Exploration and place them around the room to provide the opportunity for movement during the lesson.			

 Teacher Tips	Common Misconceptions • Students may subtract predicted from actual, resulting in incorrect signs for the residual values. • Students may add the predicted and actual values. • Students may misidentify a data point as an outlier. • Students may not recognize a data point as an outlier.	Things to Consider • This lesson is for students taking Algebra 1 Honors, as the benchmark addressed is not part of the regular Algebra 1 course. • Consider revisiting the 3.H1.1 Warm-Up during the 3.H1.4 Bring It Together component, as students may refine their answers at this point in instruction.
	Literacy and Language Routines Which One Doesn't Belong? 3.H1.1 Warm-Up provides students with four different residual plots and asks them to determine which of the four does not belong. Students are prompted to explain their rationale for deciding that one option doesn't belong, and are given opportunities to make their rationale more precise.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.7.1: Real-World Contexts This lesson is comprised solely of problems that are presented in a real-world context. Students will connect mathematical concepts to everyday experience and use models and methods to understand, represent, and solve problems related to scatter plots, residual plots, and bivariate data.
	 Differentiate Instruction	Consider taking the data sets in 3.H1.3 Exploration and placing them around the room. Have students move to each data set and discuss what information they can gather about the data based on both the scatter plot and the residual plot. Have each group rotate through each data set as they build a foundational understanding of each one for the activity.
Lesson Notes:		

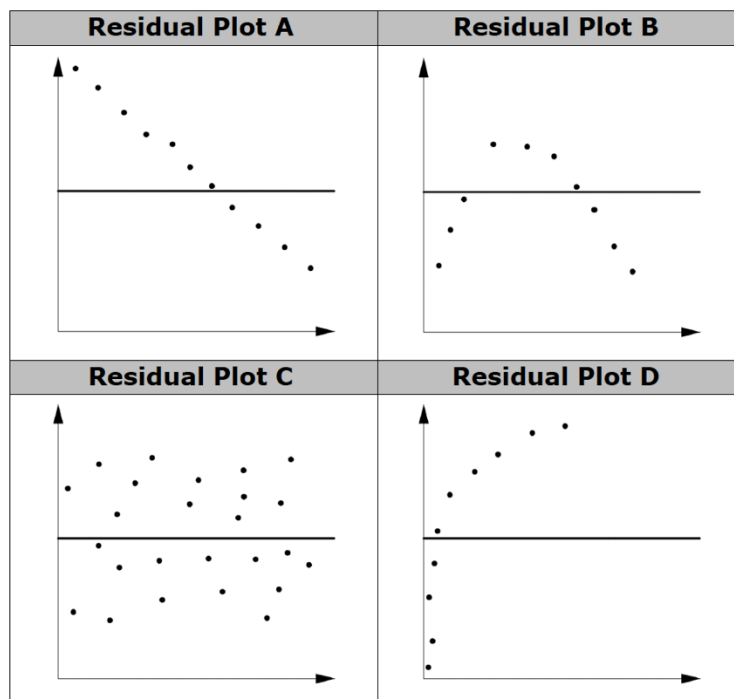
3.H1.1 Warm-Up: Which One Doesn't Belong?

Component Overview: Students will consider four residual plots to determine which one doesn't belong.



3.H1.1 Warm-Up: Which One Doesn't Belong?

Four residual plots are shown.



For this activity, an argument can be made for any of the residual plots. The goal of this activity is not for students to recognize which residual plot means the function is a good fit. It is to get them to notice patterns in the residual plots. These will come back later in this lesson.

1. Which residual plot does not belong?

Answers will vary.

2. Explain your choice.

Answers will vary.

3.H1.2 Collaboration: Creating a Residual Plot

Component Overview: Students will work together to create a residual plot for a given set of data that represents a real-world context.

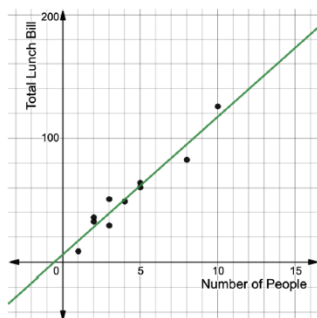


3.H1.2 Collaboration: Creating a Residual Plot

The owner of a local restaurant selected a random sample of lunch tables at her restaurant. For each table, the owner recorded the total amount of the lunch bill, before tip, and the number of people at the table.

Number of People	Total Lunch Bill
1	\$8.49
2	\$35.92
2	\$32.54
3	\$29.37
3	\$50.65
4	\$48.78
5	\$63.92
5	\$60.21
8	\$82.56
10	\$125.71

The owner's data is represented on the scatter plot with the function $B(n) = 5.82 + 11.16n$ fit to the data, where n is the number of people at a dinner party and B is the total amount of the lunch bill.



1. Work with your partner to use the linear function that models the relationship between the data and the actual data values to complete the table.



To scaffold learning of prior lessons, consider asking the following question:

- How would you determine the linear function of this data set?

3.H1.2 Collaboration: Creating a Residual Plot (continued)

Number of People	Actual Total Bill	Predicted Total Bill	Residual Values
1	\$8.49	\$16.98	-8.49
2	\$35.92	\$28.14	7.78
2	\$32.54	\$28.14	4.4
3	\$29.37	\$39.30	-9.93
3	\$50.65	\$39.30	11.35
4	\$48.78	\$50.46	-1.68
5	\$63.92	\$61.62	2.30
5	\$60.21	\$61.62	-0.41
8	\$82.56	\$95.10	-12.54
10	\$125.71	\$117.42	8.29

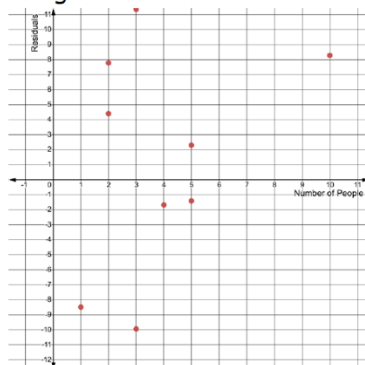
A **residual** is the difference between an actual data value and the predicted value. It is the vertical distance from the data point to the line of best fit.

$$\text{Residual} = (\text{actual } y\text{-value}) - (\text{predicted } y\text{-value})$$

2. Use your calculations to complete the table.

Number of Positive Residuals	Number of Negative Residuals	Residual with the Smallest Absolute Value	Residual with the Largest Absolute Value
5	5	-0.41	-12.54

3. Work with your partner to plot the residual values on the coordinate grid shown.



Have students check their work with other partner pairs to ensure they have the correct values. The most common error on this question is calculation errors.



To help students better understand the residual plot, consider asking the following question:

- What information do you think a residual plot would give us?

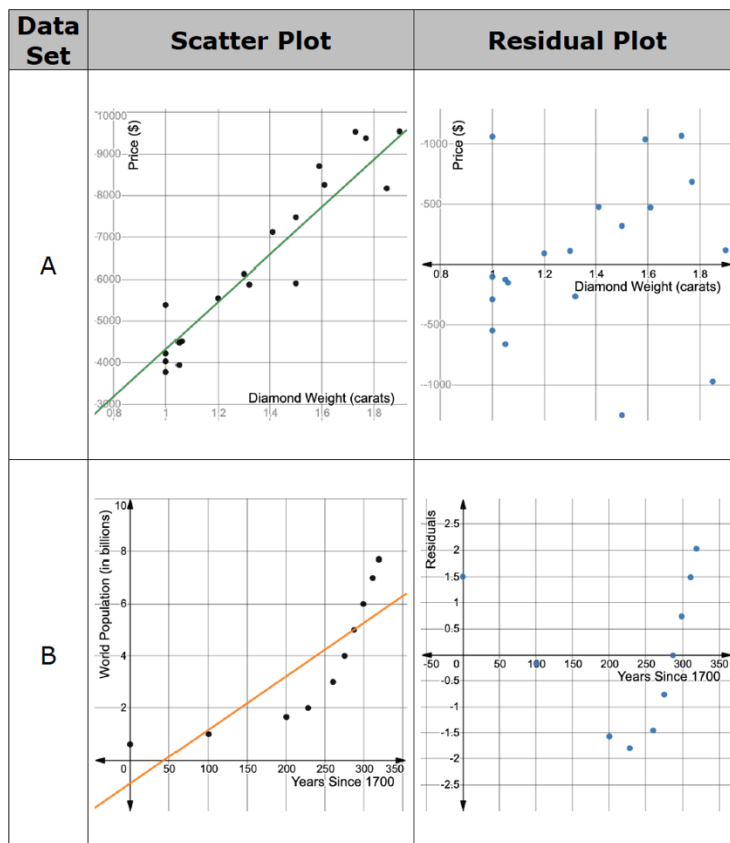
3.H1.3 Exploration: Determining the Fit of a Function by Analyzing Residuals

Component Overview: Students will work with a partner to explore how to determine the fit of a function by analyzing residuals given three sets of data.

3.H1.3 Exploration: Determining the Fit of a Function by Analyzing Residuals

Work with your partner to complete the following.

Three scatter plots are shown, each fit with a linear function. The residual plots comparing the actual data values on the scatter plot to the predicted values using the linear functions are also shown for each set of bivariate data.



Consider taking the data sets in this activity and placing them around the room. Have students move to each data set and discuss what information they can gather about the data based on both the scatter plot and the residual plot. Have each group rotate through each data set as they develop a foundational understanding of each one for the activity.

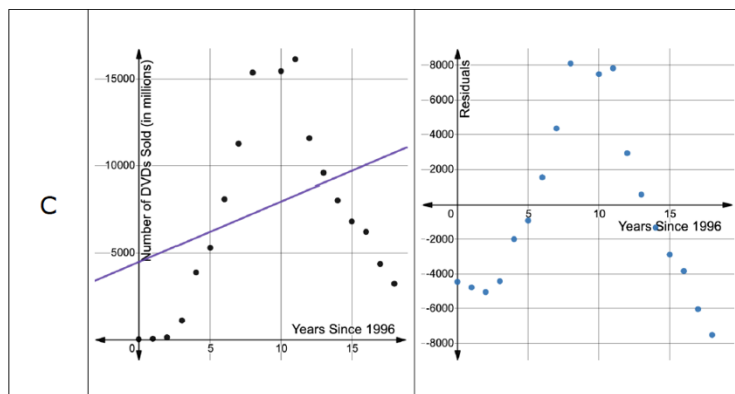


This exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group 3.H1.4 Bring It Together.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

3.H1.3 Exploration: Determining the Fit of a Function by Analyzing Residuals (continued)



1. Work with your partner to explain which scatter plot appears to show the strongest linear relationship.
Answers will vary. Scatter Plot A appears to show the stronger linear relationship because the trend of the points best match with the line of fit.
2. Describe how the residual plot for the linear function fit to the scatter plot selected in the previous question appears.
Answers will vary. The residual plot for Scatter Plot A was the most random and above and below 0.
3. Which phrase best describes the residual plots for the functions that model relationships that do not appear to be linear?
 - The residual plot includes points above and below the x -axis that appear random and not following a pattern.
 - The residual plot includes points that appear to follow a pattern.



At the conclusion of this activity, consider engaging the class in a whole-group discussion on their answers to these three questions. This is an opportunity to clear up any potential misconceptions from their work, prior to bringing everything together in the next activity.

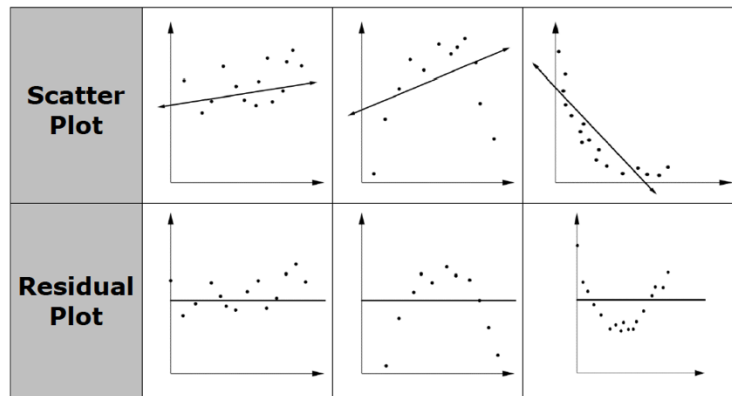
3.H1.4 Bring It Together: Determining the Fit of a Function by Analyzing Residuals

Component Overview: Students will bring together what they've learned from the prior two activities to understand how to determine the fit of a function by analyzing residuals.



3.H1.4 Bring It Together: Determining the Fit of a Function by Analyzing Residuals

Three scatter plots with a linear function fit to the data and their corresponding residual plots are shown.



The residual plot can be used to determine whether or not the function is a good fit.

1. Complete the statements.

If the function is a good fit for modeling the relationship between variables, the residual plot should include

- ☐ points that follow a pattern
 - ☒ points that are randomly scattered
- and

- ☒ a balance of points above and below the x -axis.
- ☐ most of the points on one side of the x -axis.

The better the function fits the data, the

- ☒ closer
 - ☐ farther away
- the residual values will be
- ☒ to
 - ☐ from

the x -axis.



When reviewing the information in question 1, use language that connects the formal information back to what they did in the prior activities. Students should understand that this is the formal conclusion of the exploration they just completed.



To check students' understanding of why the closer the residual values are to the x -axis, the stronger the fit, consider asking the following questions:

- Why do you think residual values closer to the x -axis represent a stronger fit?
- How is this shown on the scatter plot?

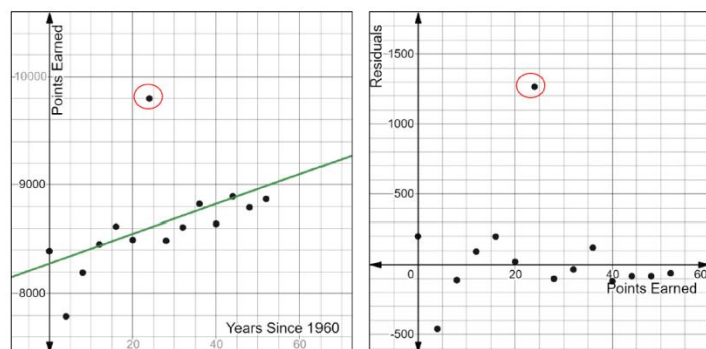
3.H1.4 Bring It Together: Determining the Fit of a Function by Analyzing Residuals (continued)

Residual plots can also be used to identify outliers in the data set.



An **outlier** is a value that is much higher or much lower than the other values in a set of data.

2. The scatter plot with a linear function fit to the data and its corresponding residual plot are shown. The scatter plot represents the relationship between the number of points earned by men's decathlon Olympic gold medalists and years since 1960.



- a. Circle the point on the scatter plot that appears to be an outlier and its corresponding residual value on the residual plot.

Answer on graph above.

- b. Complete the following statement.

In a set of bivariate numerical data, the residual of

an outlier has the ☐ smallest ☒ greatest absolute value

when compared to other residual values.



Why do you think that an outlier would have the greatest absolute value compared to other residuals?

3.H1.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door on analyzing residuals and determining if an outlier appears to exist.

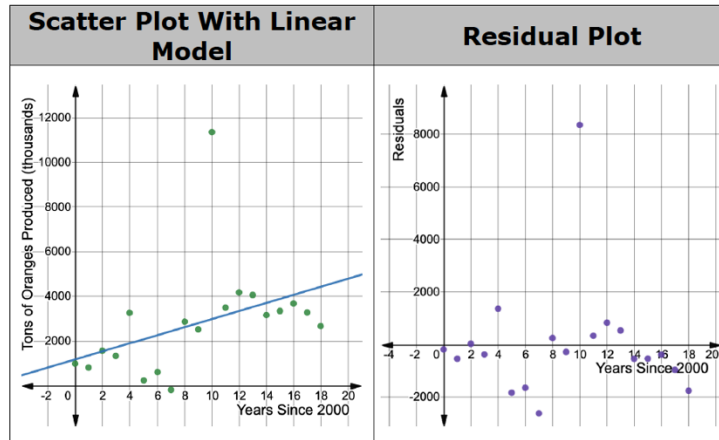


3.H1.5 Wrap-Up: Ticket Out the Door

One of the many things that Florida is known for are its oranges. Oranges, however, are grown and harvested in multiple states across the U.S. Below is a scatter plot representing U.S. orange production since 2000. A linear function is fit to the data, and the corresponding residual plot is shown.



Student performance on this activity can provide data on whether students understood the content of this lesson.



1. Explain whether or not the linear function is a good fit for this data set.

Answers will vary. The linear function is a good fit for the data because in the residual plot, the points are all randomly mixed above and below the x-axis.

2. Explain whether or not there appears to be an outlier in the data set using the residual plot.

Answers will vary. Based on the residual plot, there appears to be an outlier in the data because there is a point with a residual value whose absolute value is much greater than that of all the other residuals.